Sequence to Sequence (seq2seq) & Attention

Ling 282/482: Deep Learning for Computational Linguistics

C.M. Downey

Fall 2026

Sequence-to-sequence problems

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- seq2seq: input a sequence and output a **different sequence**

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    - Semantic as well as syntactic

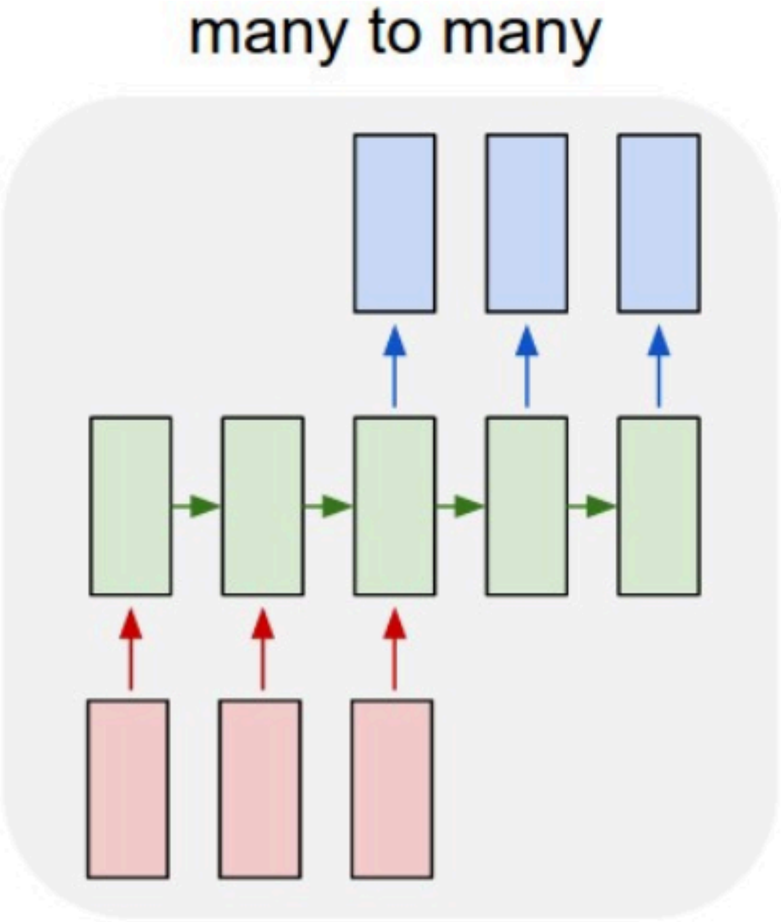
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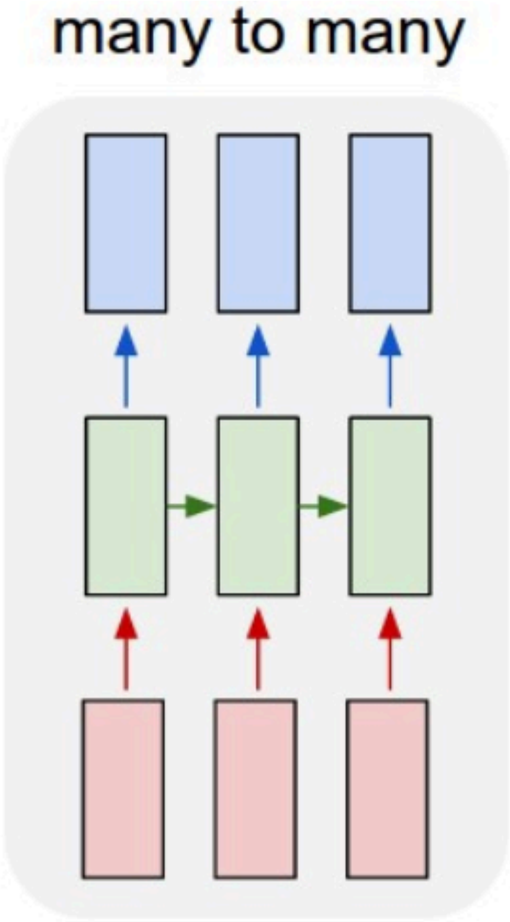
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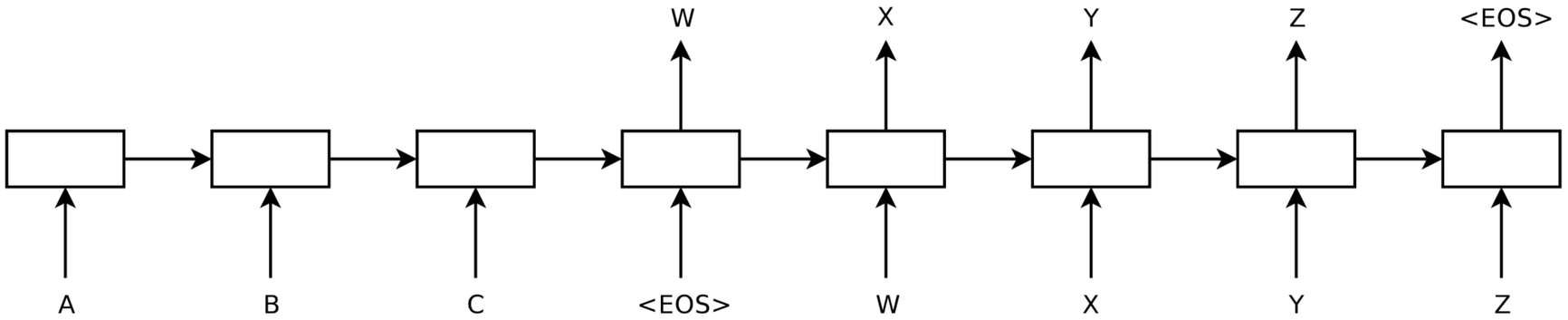
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  - **Summarization** and **Question Answering**
- **Not the same** as *tagging*, which assigns a label to each position in a given sequence

seq2seq vs Tagging



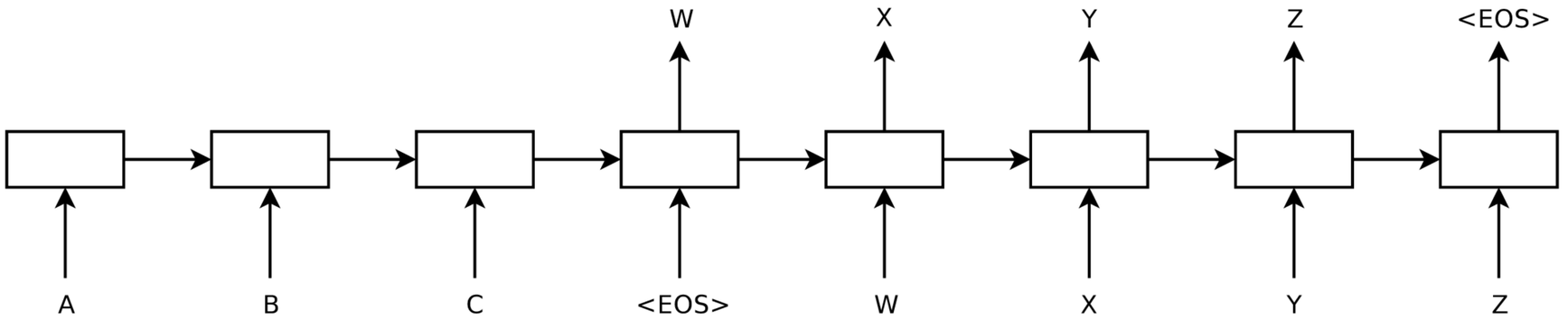


seq2seq architecture



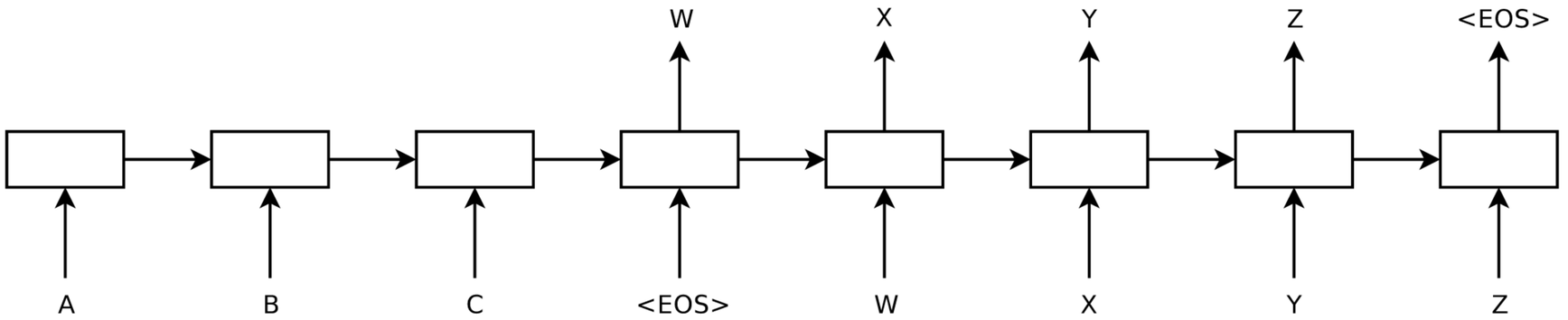
Sutskever et al 2013

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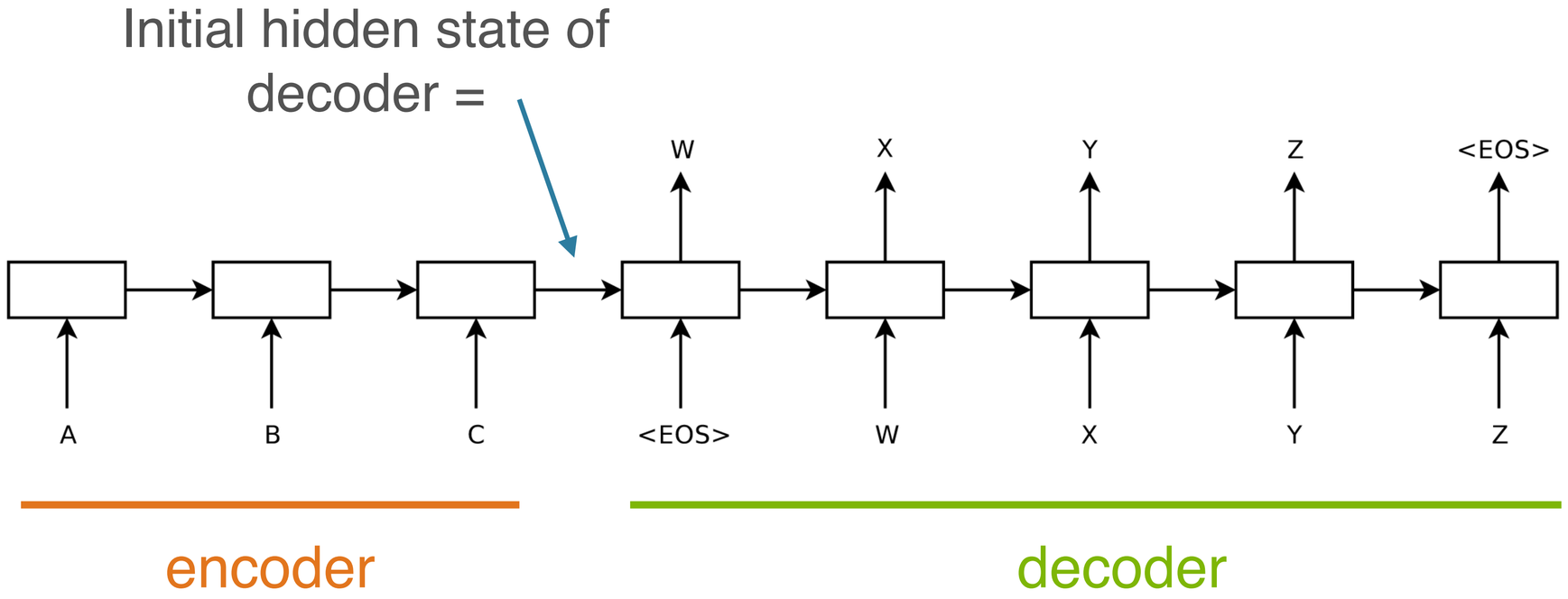
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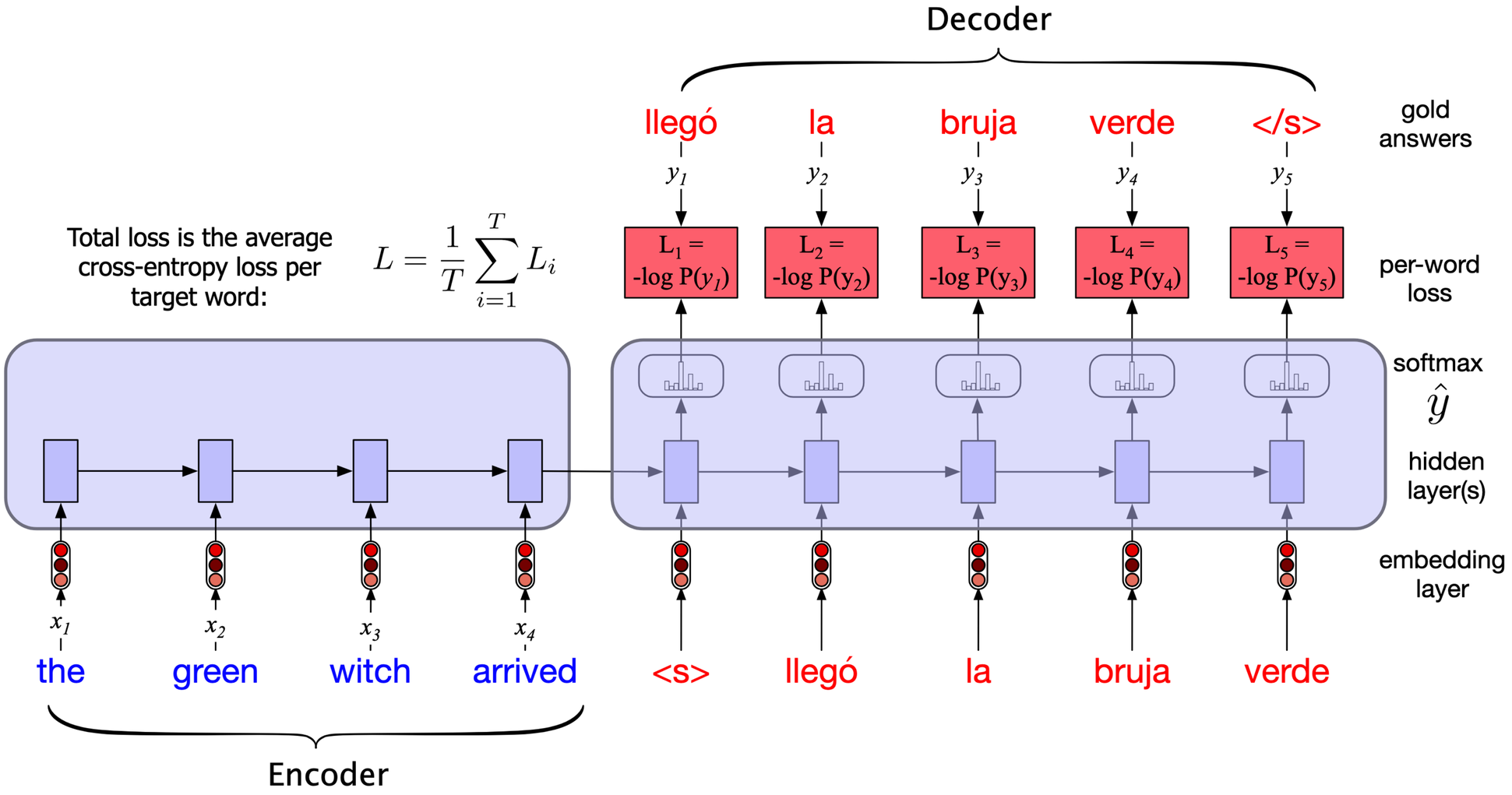
seq2seq architecture

- Two components
  - **Encoder**
    - Input sequence —> vector representation (“context” vector)
  - **Decoder**
    - Vector (“context” vector) —> Output sequence

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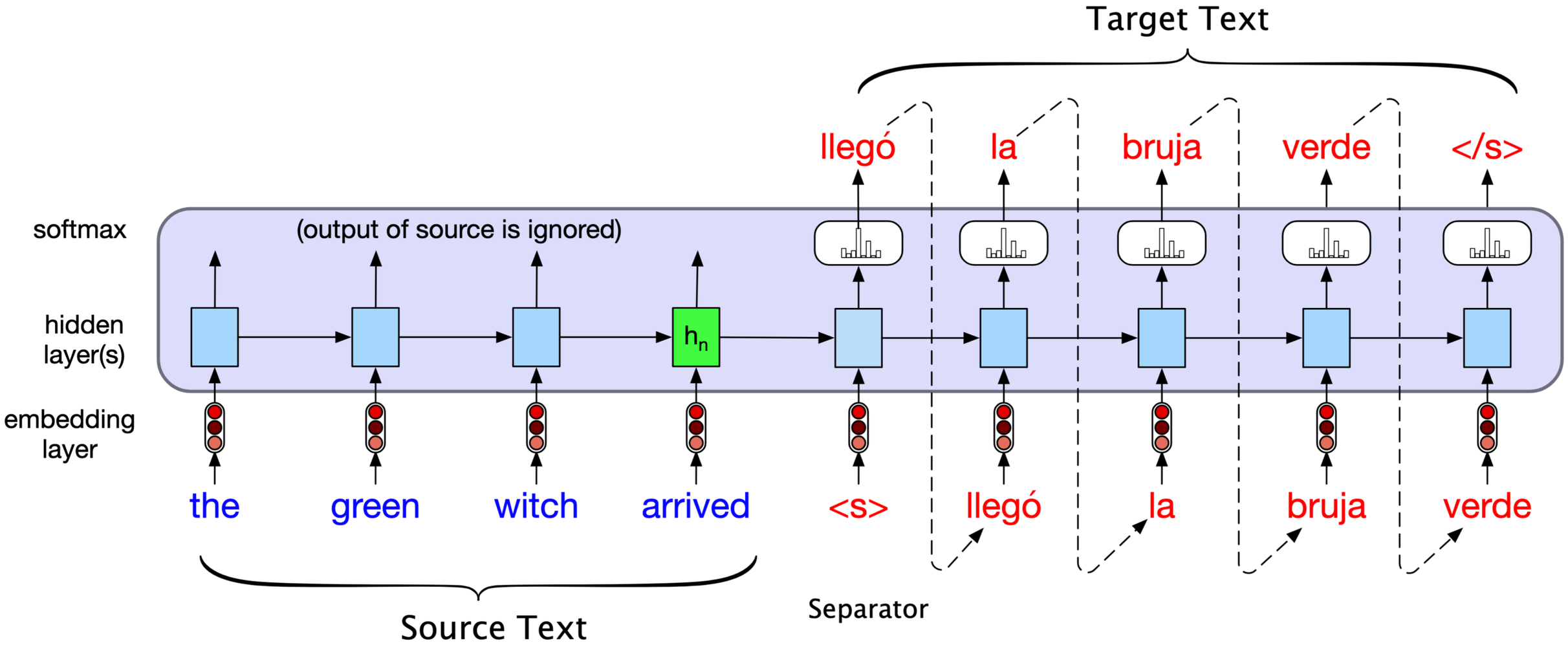
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- High-level **API**
  - Encoder/decoder can be **different architectures** (LSTM, GRU, Transformer, convolutional, …)

Training an encoder-decoder RNN



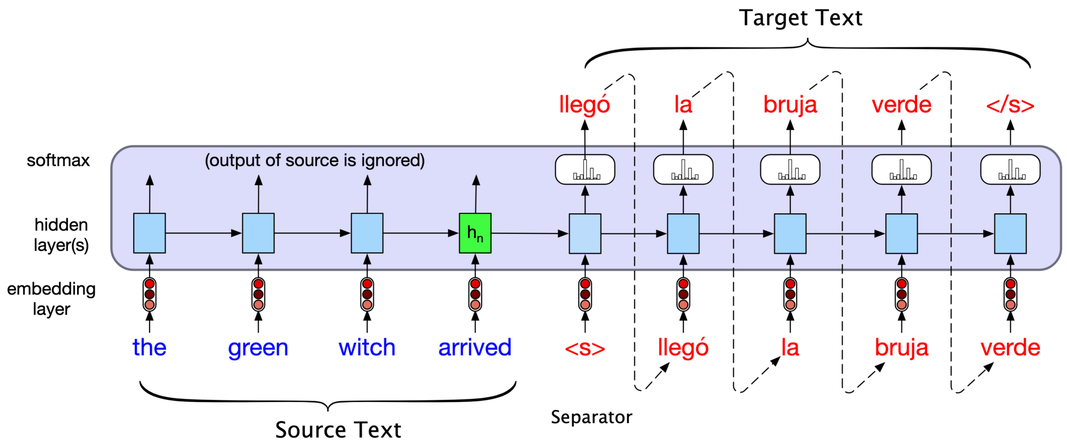
JM 13.7

Inference / Generation



Conditional LM

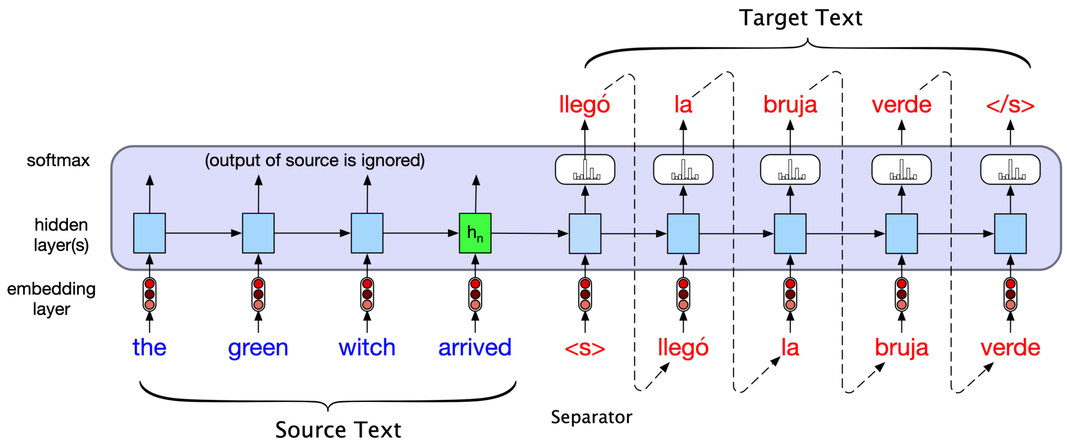
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Conditional LM

- Effectively, a seq2seq model is a **Conditional Language Model**

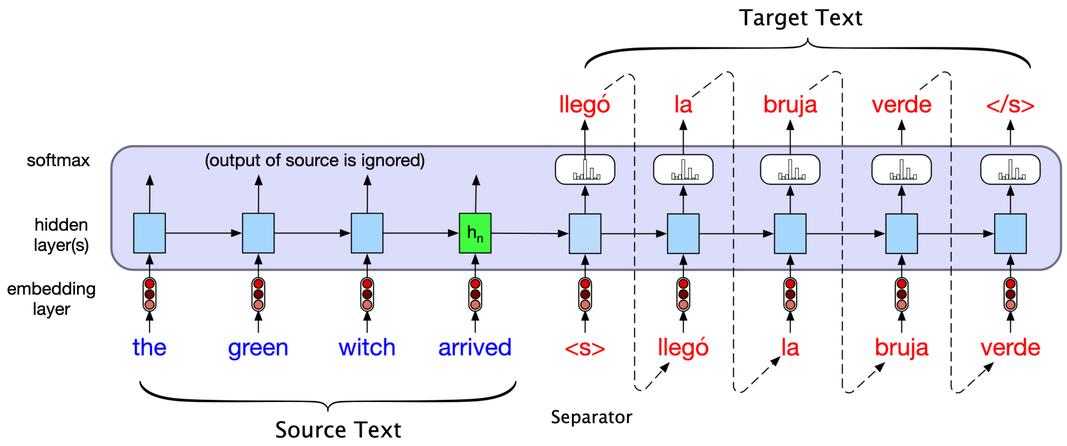
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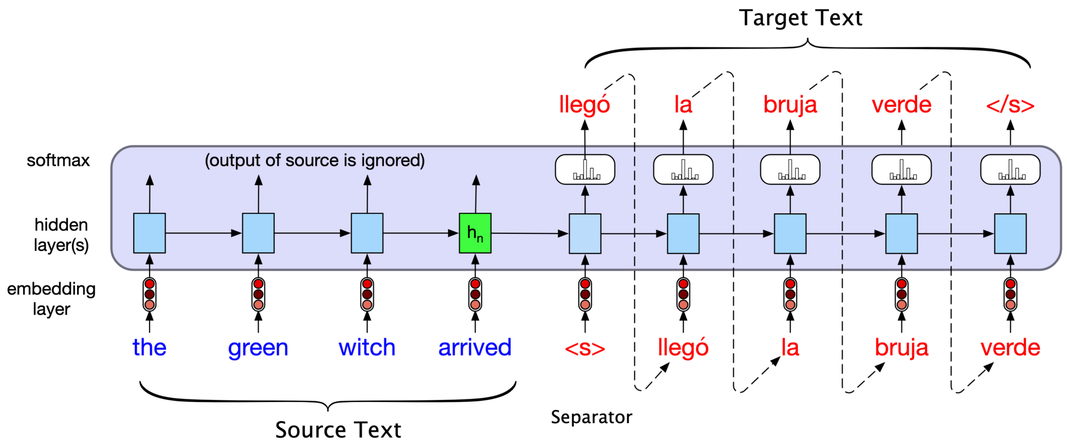
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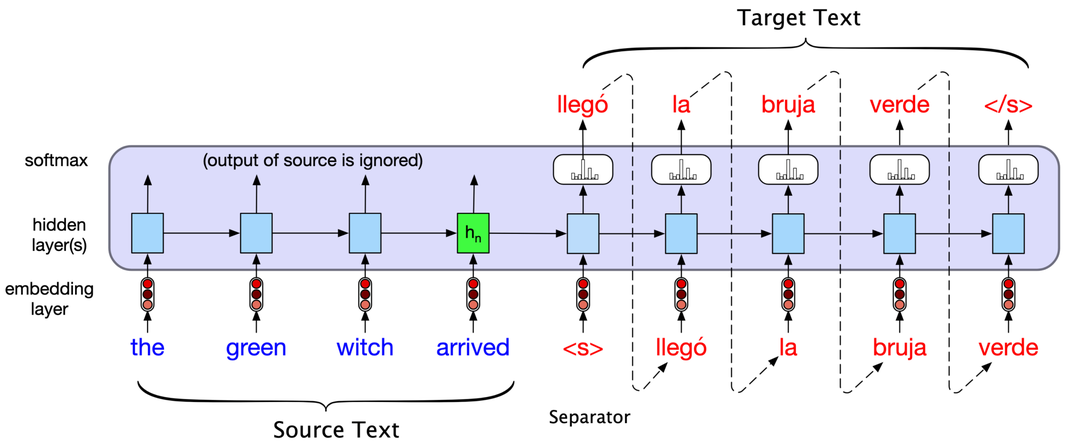
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Conditional LM

- Effectively, a seq2seq model is a **Conditional Language Model**
- LMs like we have seen, but **conditioned on the input**
- LMs were already conditioned on the **output sequence prefix**
- Each step of the output is conditioned on the **whole** of the input

$$P(y \mid x) = \prod_{i=1}^{|y|} P(y_i \mid x, y_{<i})$$



Sequence Alignment

Statistical Machine Translation (90s-2010s)

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- Goal: find best translation *y* (e.g. English) of source sentence *x* (e.g. French)

$$\arg\max_{y} P(y \mid x)$$

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- Use Bayes' Rule to decompose into two components:

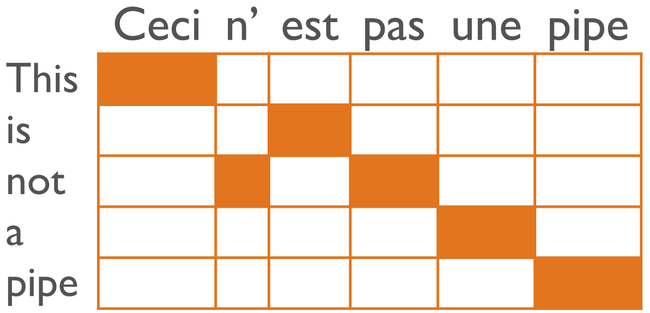
$$\arg\max_{y} P(x \mid y)P(y)$$

  - Core **translation model: P(xly)**
  - "Pure" **Language Model P(y)**: produce good / fluent target language text (e.g. English)

Alignment

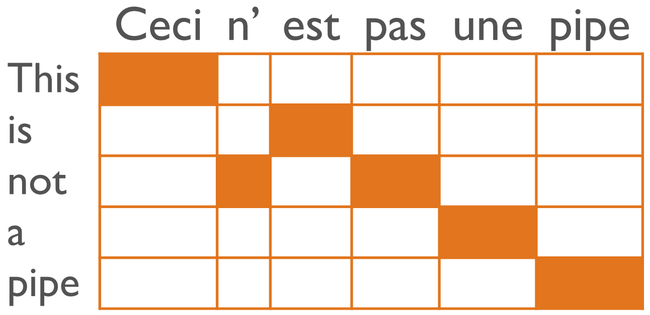
- Most SMT systems modeled **alignment** between sequences
  - **Correspondence between words/phrases** in source and target sentence
  - Useful since languages have **very different word orders**
- Add alignment as a latent variable:

$$P(x, a \mid y)$$



Alignment, example

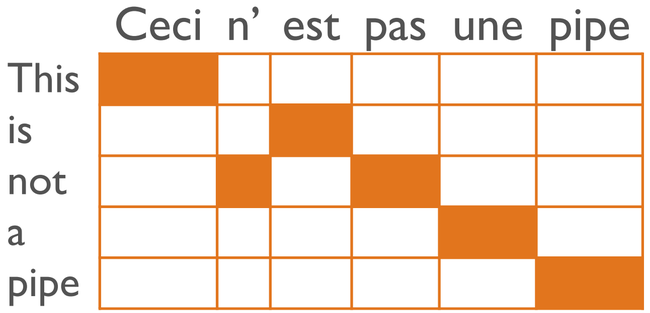




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Ceci n’ est pas une pipe

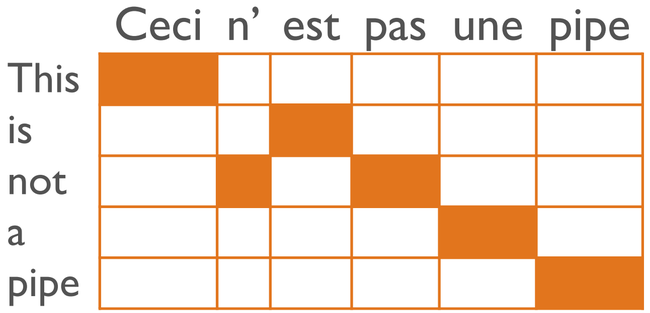


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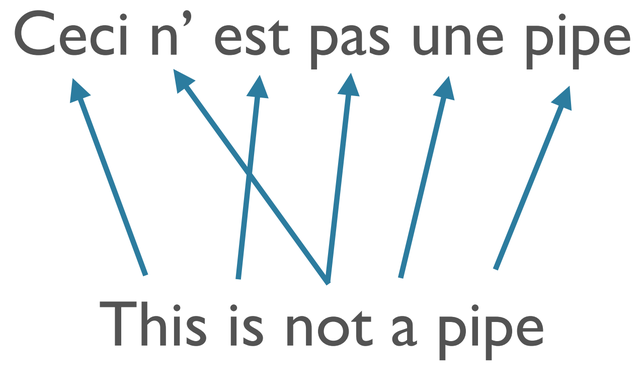
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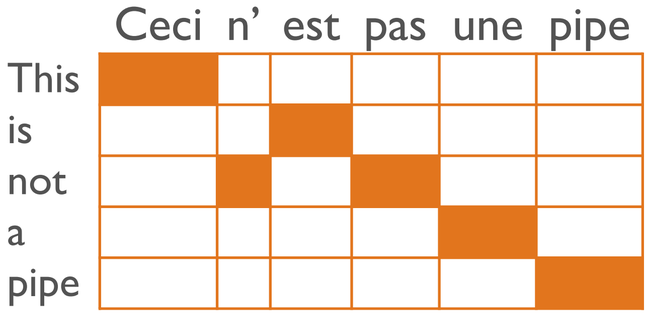
This is not a pipe



Alignment, example







SMT Difficulties

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- **Key features** for determining alignment
  - Probability of **word-pairs** aligning (using a **lexicon / bilingual dictionary**)
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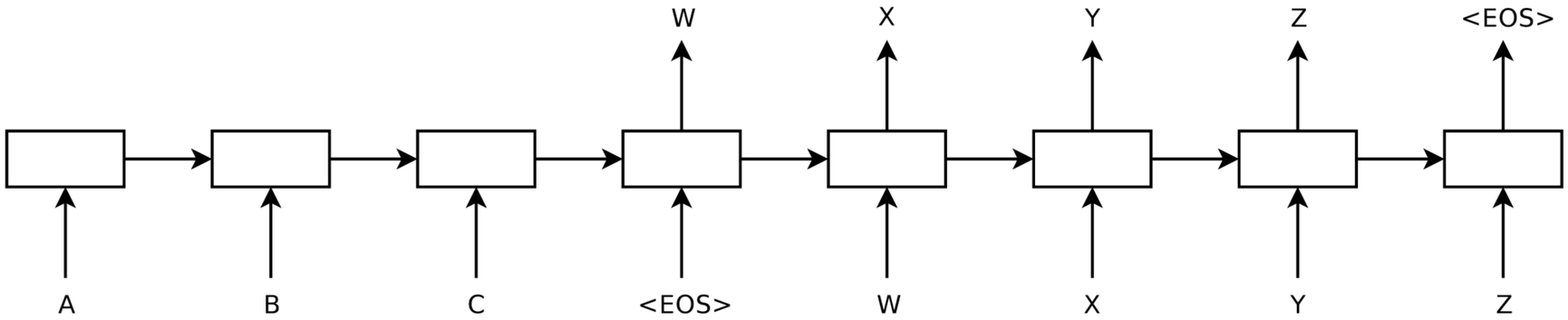
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  - Most of the above are **language-pair-specific**, have to be repeated
- MT was one of the **first major success stories** of neural methods in NLP:
  - End-to-end systems, “language-agnostic” models, equal/better performance

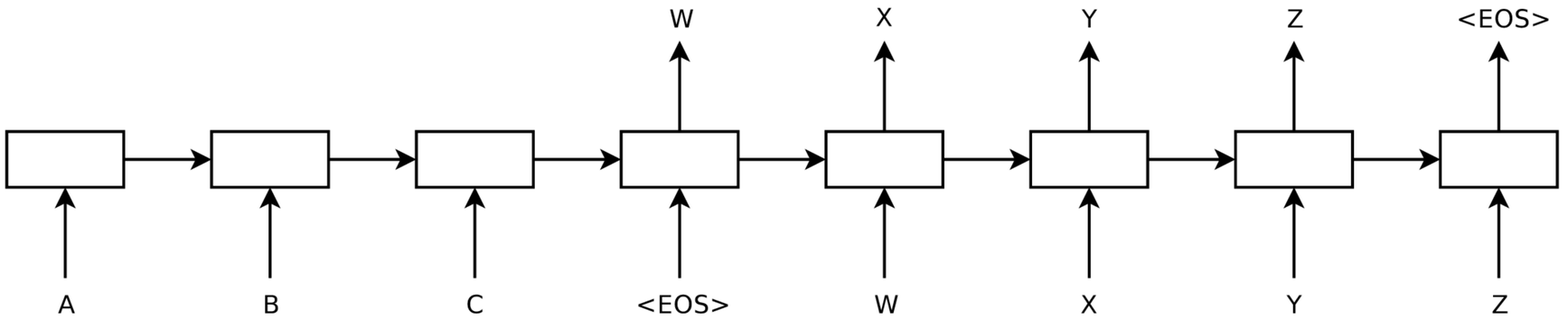
Attention

seq2seq architecture: problem



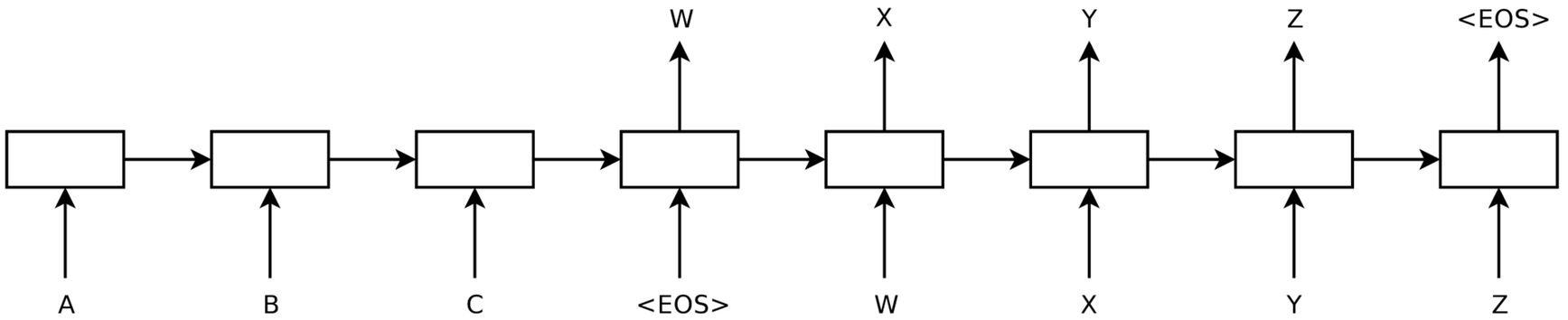
Sutskever et al 2013

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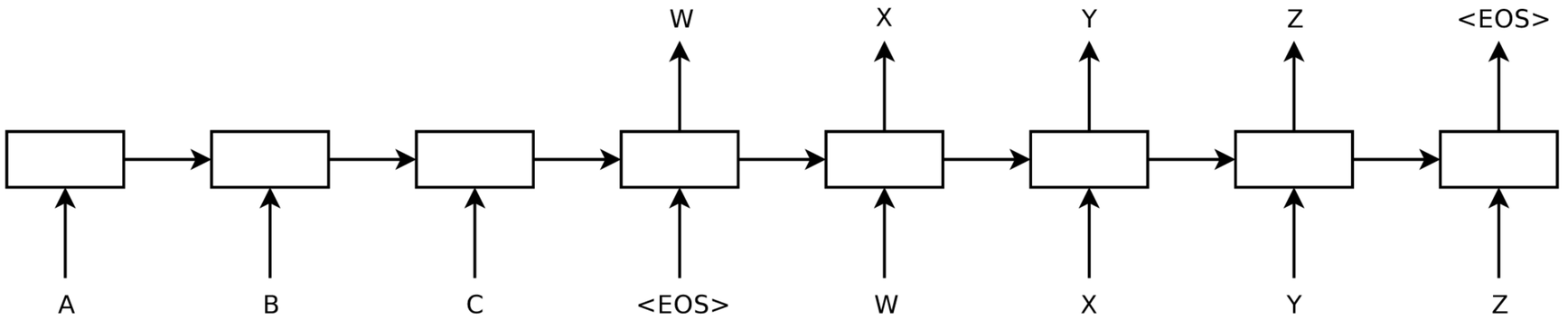
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Sutskever et al 2013

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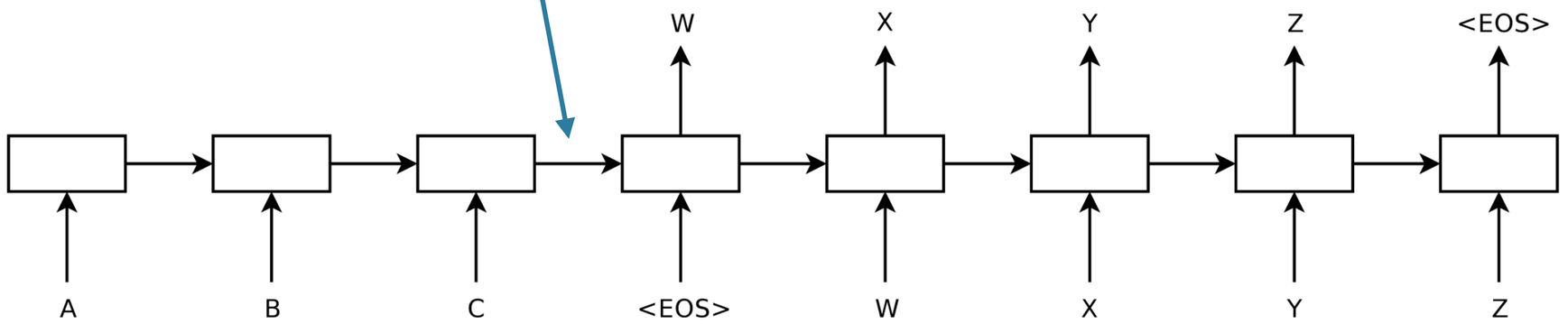


Sutskever et al 2013

seq2seq architecture: problem

Decoder can only see info in this

Mooney 2014: “You can't cram the meaning of a whole %&!$# sentence into a single $&!#* vector!”



Sutskever et al 2013

NEURAL MACHINE TRANSLATION BY JOINTLY LEARNING TO ALIGN AND TRANSLATE

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ABSTRACT

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source

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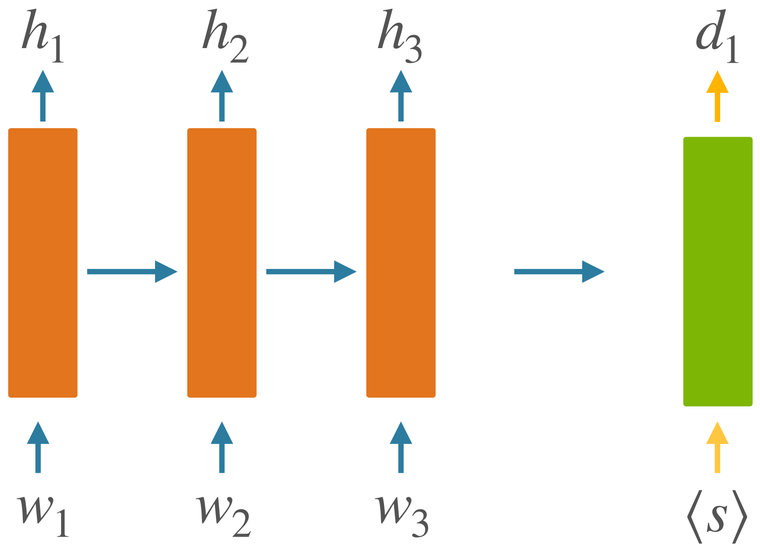
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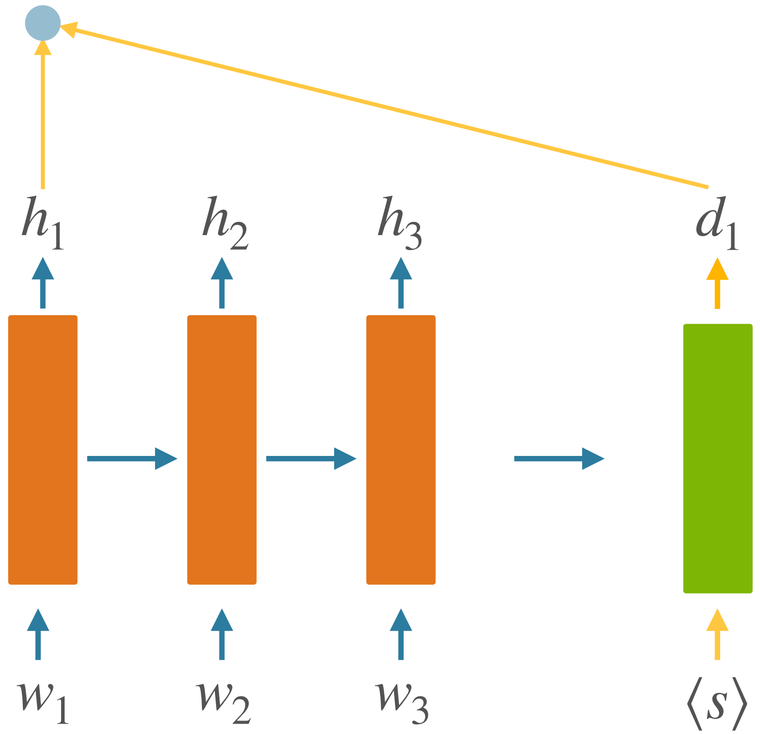
source

Adding Attention



Badhanau et al 2014
Luong et al 2015

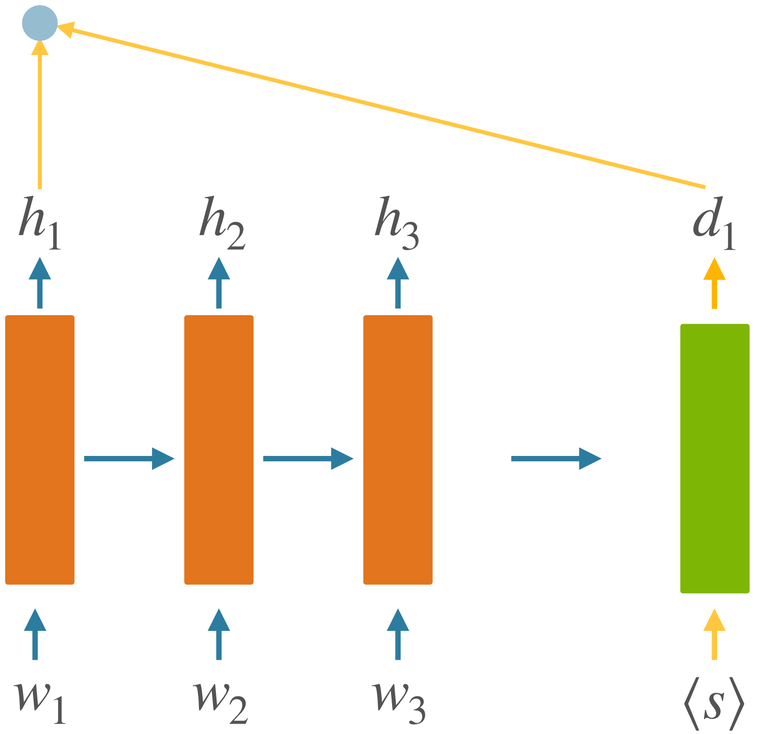
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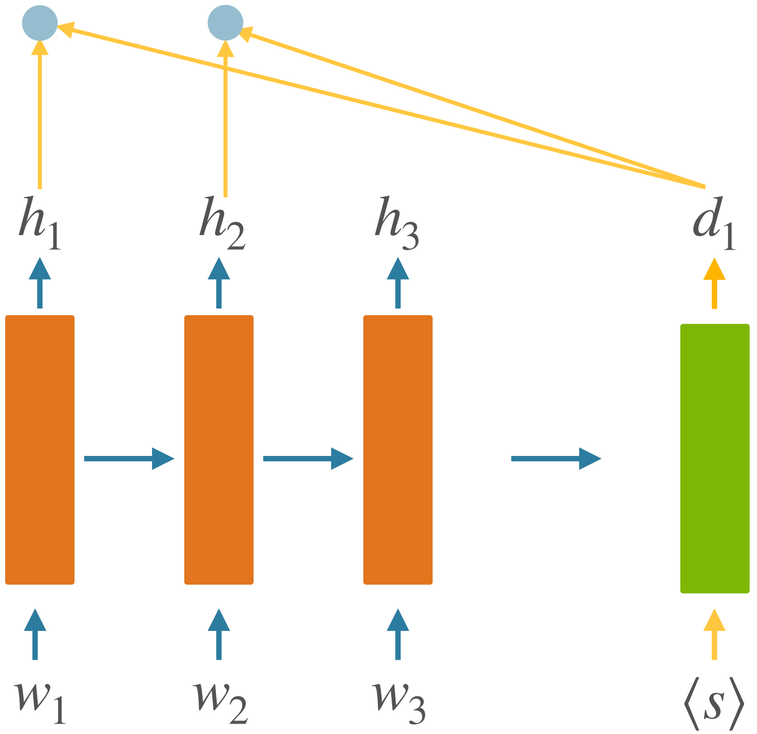
$\alpha_{ij} = a(h_j, d_i)$
(dot product usually)



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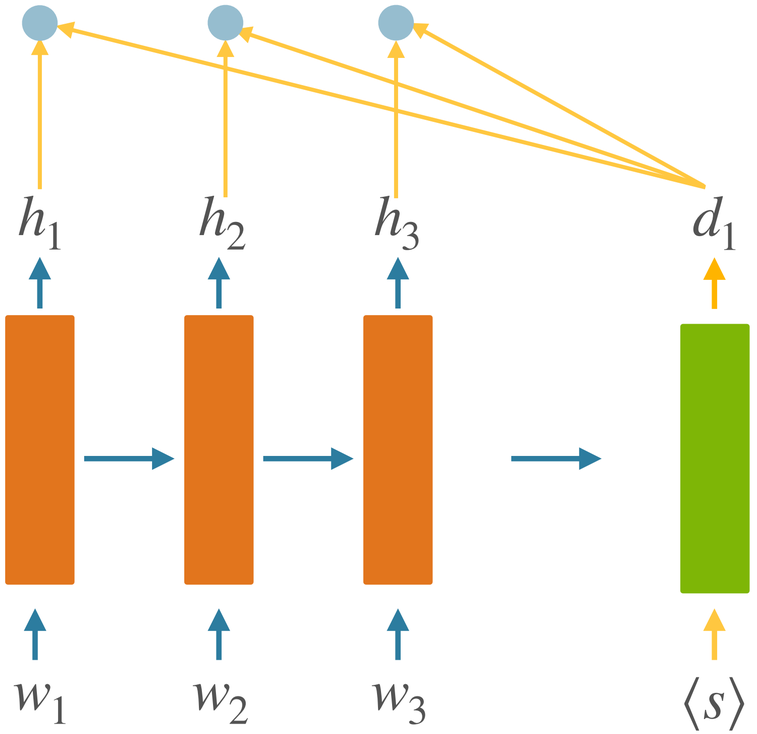
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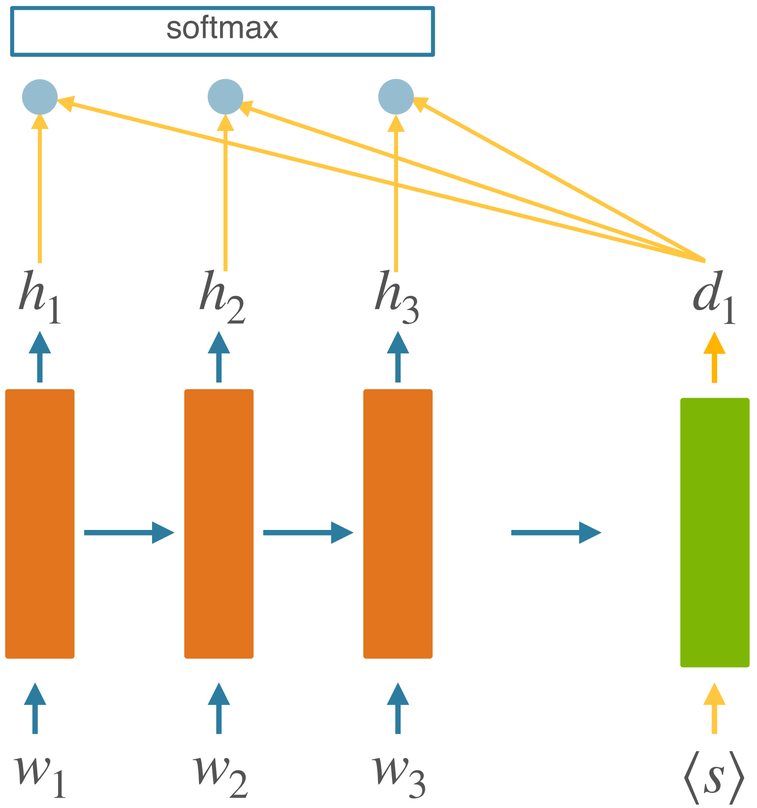


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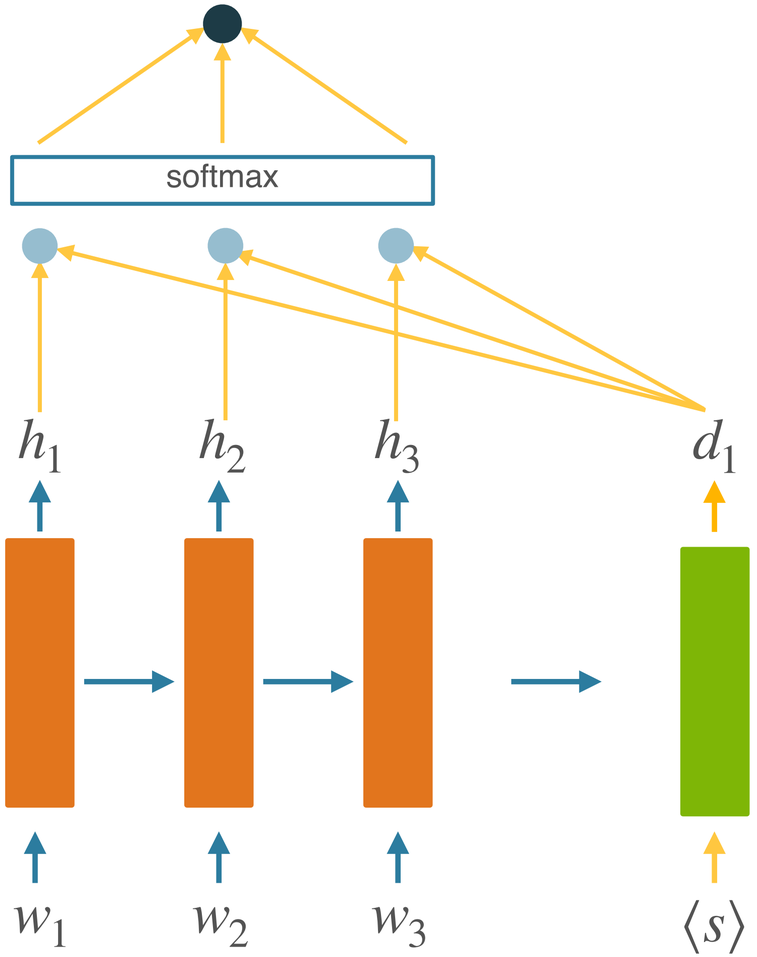
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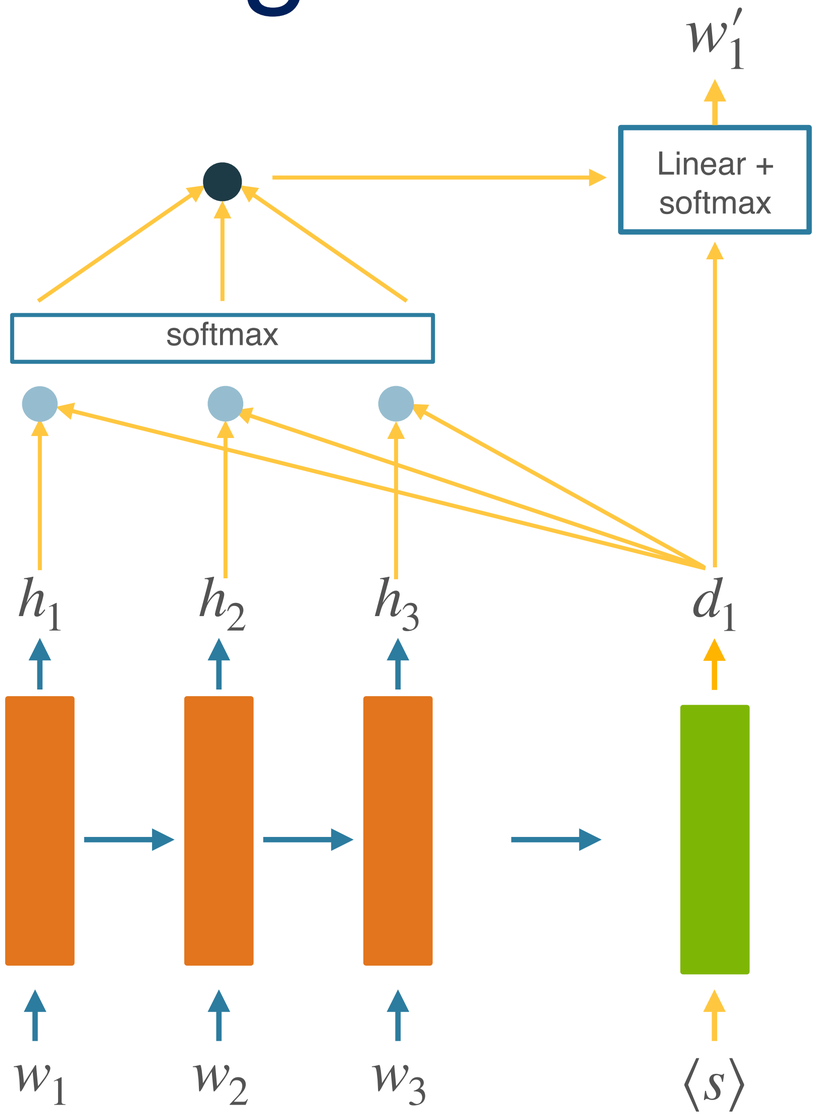
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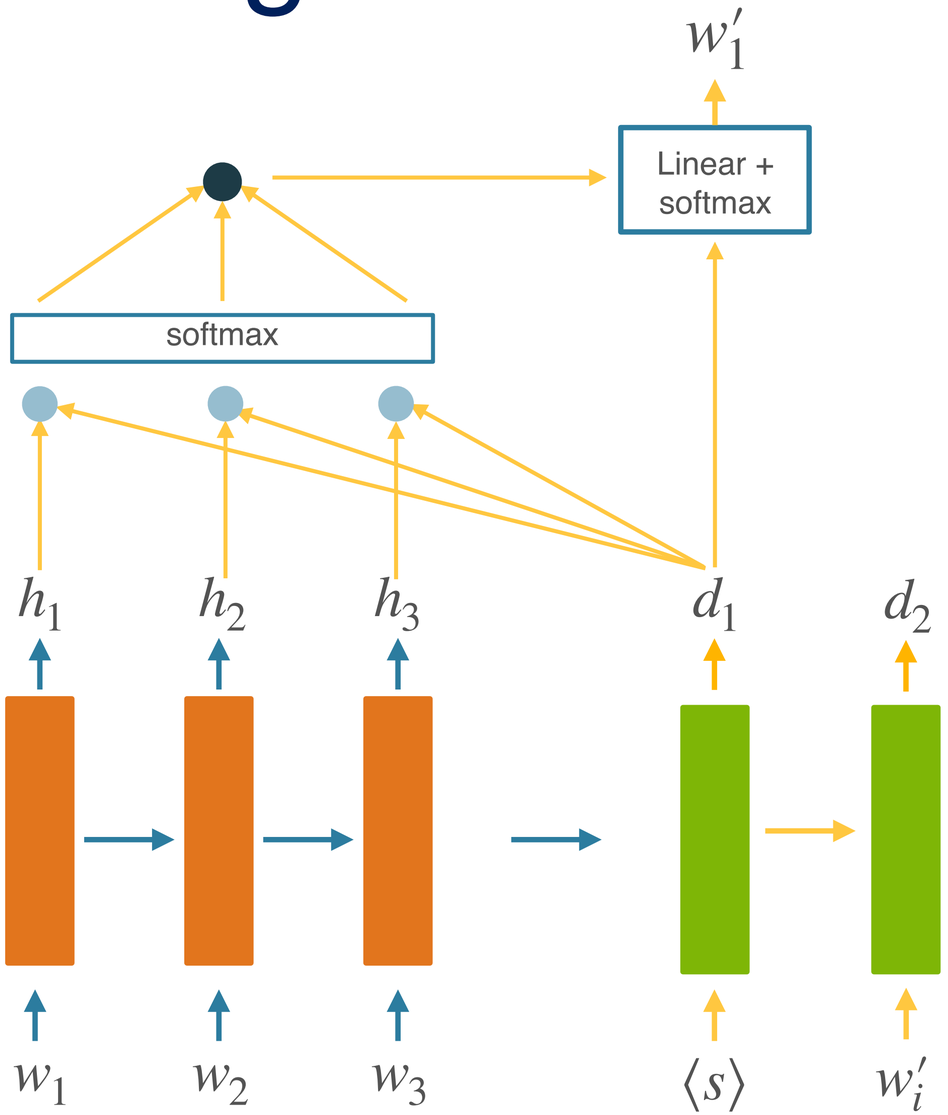
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- In the previous example: **encoder hidden states** played **both** the keys and the values roles

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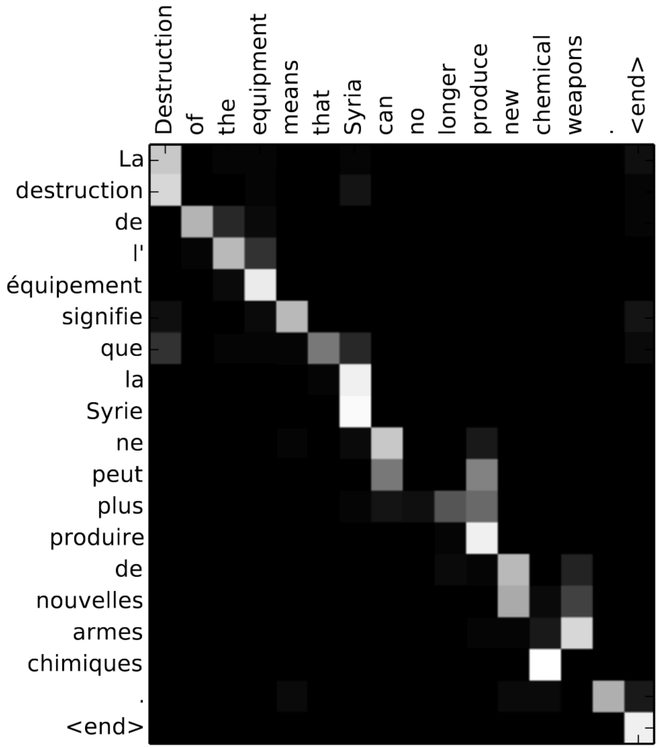
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  - By **“solving” the bottleneck issue**

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- Aids **interpretability (maybe)**

Why attention?

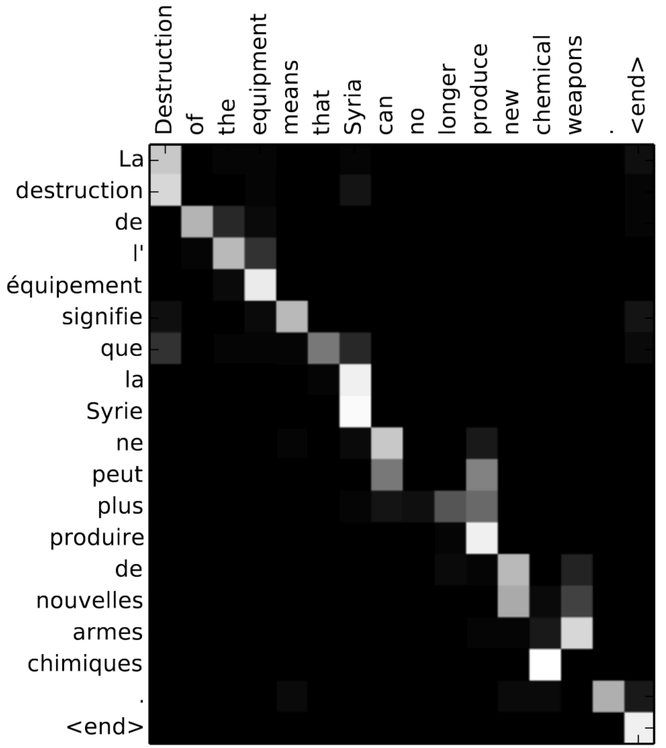
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  - By **“solving” the bottleneck issue**
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Badhanau et al 2014

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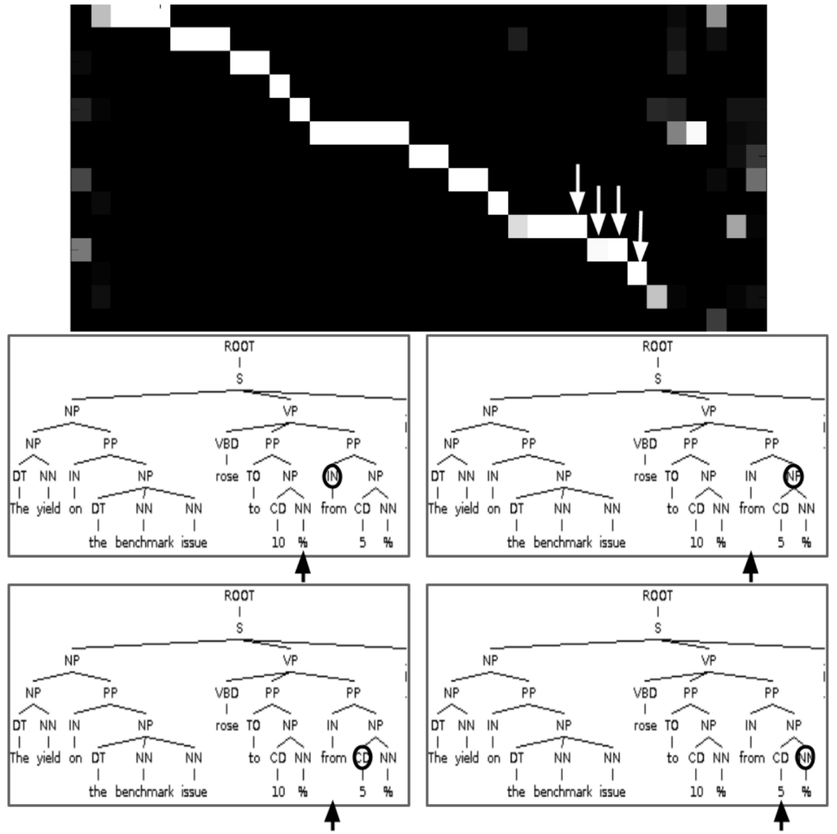
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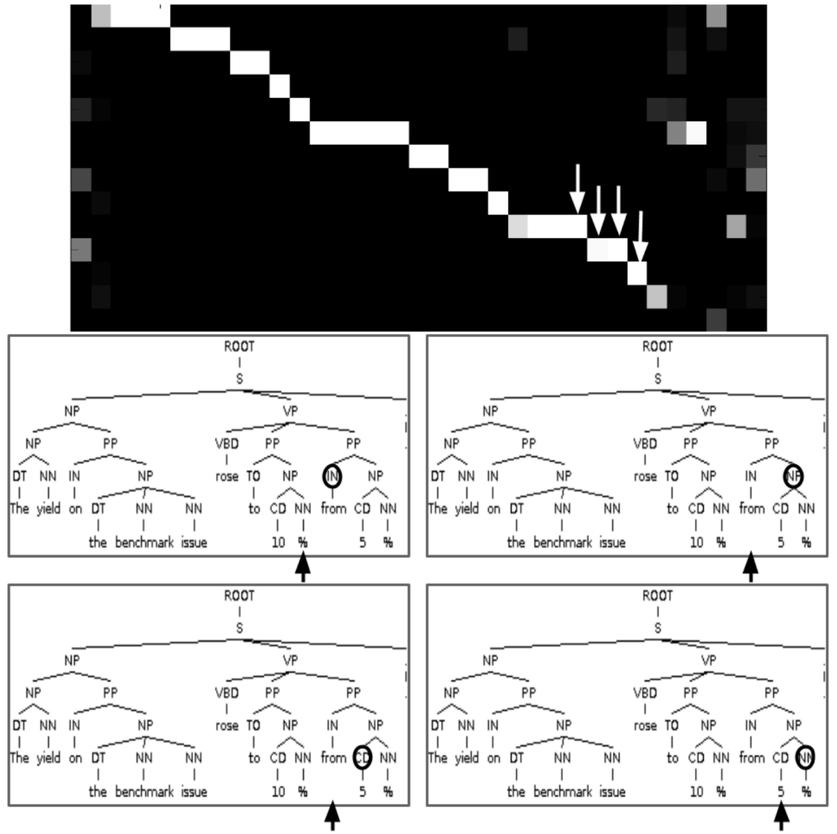
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Vinyals et al 2015

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- Conceptually, let the model **learn to align** representations
  - “Soft” alignment, just like gates = “soft” masks



Vinyals et al 2015